ZEDPATH

Team Kestrel · IntelliCon '26 Buildathon

ZP-DOC-01

Vision and Scope

Business requirements, product vision, scope boundaries and business context

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Document control

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Revision history

Version	Date	Author	Description of change
0.9	5 Oct 2026	Team Kestrel	First issue, derived from the Gate 1 Problem and Proof deck; submitted for approval
1.0	5 Oct 2026	B.M.P Banneka	Approved by the product owner; baselined
2.0 (draft)	11 Oct 2026	Team Kestrel	Under revision for change requests CR-001 to CR-006 and the video review of 11 October 2026 (all approved by the product owner): change-request summary added; features FE-1, FE-4 to FE-7 restated; release scope recomputed from ZP-DOC-02 (R1 57 stories, 189 points; R2 10 stories, 31 points); limitations updated (account required; Android app exists; reminders on the phone); RK-02 and RK-07 mitigations updated; RK-08 added; market figures unchanged. Scope change, so the next approved version is 2.0. Version 1.0 of 5 October 2026 stays in force until the product owner approves this revision

Review and approval

This document takes effect when the approver signs below. Until then its status is as shown on the cover.

Role	Name	Signature	Date
Reviewer	Claude (AI coding partner), consistency and traceability review		
Approver	B.M.P Banneka, Product Owner	v1.0 approved in writing (project channel); this revision awaits approval	5 Oct 2026 (v1.0)

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1 Introduction

1.1 Purpose

This document states **why** ZedPath is being built, **what** it will and will not do, and **how** its success will be judged. It is the parent of every other ZedPath document: user stories (ZP-DOC-02) refine its features, and the requirements specification (ZP-DOC-03) makes them precise.

1.2 Scope

It covers the business requirements, the product vision, the scope of the first two releases, and the business context (stakeholders, priorities, risks and deployment constraints). Technical design is out of scope.

1.3 Intended audience

The product owner, buildathon judges and mentors, and anyone deciding whether ZedPath is worth building.

1.4 Relationship to other documents

Table 1: Position of this document in the ZedPath document set

Document	Relationship
Gate 1 Problem and Proof deck	Source. Evidence, market sizing and the three-week plan come from it
ZP-DOC-02 User Stories	Child. Each feature FE-n below is refined into the stories of epic EPn
ZP-DOC-03 Software Requirements Specification	Child. Business objectives become measurable quality requirements
ZP-DOC-06 Software Architecture	Child. The deployment constraints below bind the architecture

1.5 Changes since version 1.0

After v1.0 was approved on 5 October 2026, the product owner approved six change requests and the changes from a video review of the live app. Items they changed are marked *Revised* below; ZP-DOC-02 and ZP-DOC-03 hold the detail.

Table 2: Change requests that affect the vision and scope

CR	Date	What changed in scope
CR-001	5 October 2026	Student accounts (username and password, key derived on the phone, recovery code); the profile follows the student across devices; delete account
CR-002	11 October 2026	Journey with reminders (phone notifications in the Android app, calendar file elsewhere); Alerts for Gazette and intake notices, checked on the phone; course and A/L-batch groups with verified seniors and moderation; navigation Paths, My list, Journey, Groups, Ask
CR-003	11 October 2026	An account is **required** at the start, with full name, email, Sri Lankan mobile and date of birth, each with its purpose and recorded consent (Personal Data Protection Act, No. 9 of 2022)
CR-004	11 October 2026	Ask answers as a card: verdict, sources, next steps from ZedPath's data, follow-ups and feedback

CR	Date	What changed in scope
Video review	11 October 2026	Sinhala and Tamil interface; search; Compare of up to three courses; chart values; achievements used in Ask; reminder settings; "Sit the A/L again" path removed; selecting several courses at once
CR-005	11 October 2026	Group rules before taking part; no phone numbers or emails in messages; popular groups; labelled sample conversations
CR-006	11 October 2026	Group rules once per account; group info page; preset avatars; the student ticks Journey steps within their windows; follow courses and notices; Journey saved on the account; Alerts become the Journey's *Open now* tab

2 Business requirements

2.1 Background

Every year about 275,000 students sit the G.C.E. Advanced Level (A/L) examination in Sri Lanka. In 2025, **281,810** sat the examination, **176,527** qualified for university admission, and there were **42,937** state-university places for the 2025/26 intake [1][2].

Within a few weeks of results, each student must decide. Students aiming for a state university must choose and rank up to 125 course preferences under rules published in a handbook of more than 200 pages that changes every year, inside a three-week application window. Because places are limited, most also need a backup: a private or non-state degree, study abroad, a government job examination announced in the Gazette, a vocational course, or sitting the examination again. Each path has its own rules, calendar and website.

Gate 1 research (six interviews, E1 to E6, and a hands-on test of four existing tools, T1) found one pattern: students decide mostly on hearsay from tuition teachers, seniors and social-media groups. A badly ordered preference list, an ineligible choice, a missed deadline or a path never heard of can cost a year. The National Education Commission reached the same conclusion: most students and parents are unaware of alternative education and career options [3].

2.2 Business opportunity

No existing tool combines **the student's own results**, **every path they can reach** and **dated deadlines** in one place. The tools tested (findmydegree.lk, Unicompass, induwara.lk, ThuSh LK) compare a Z-score with cut-offs, but none checks subject and grade rules, preference order or deadlines, and none covers routes outside the University Grants Commission (UGC) [4].

Two developments make a better tool feasible now:

1. **AI that reads documents.** Language models can turn a 200-page handbook and weekly Gazette PDFs into structured, page-cited rules, which previously needed manual re-coding every year.
2. **A free global edge platform.** Cloudflare's free plan provides hosting, database, storage, scheduled jobs, queues and AI inference at zero cost, close to users in Sri Lanka.

2.3 Business objectives

Table 3: Business objectives and how each is measured

ID	Objective	Measure and target
BO-1	Reach students in the first admission cycle	10,000 students use ZedPath in year one (serviceable obtainable market)
BO-2	Make every fact checkable	100% of displayed cut-offs, rules, dates and fees link to an official source
BO-3	Let a student plan alone	A first-time user goes from entering results to a checked preference list in under 10 minutes without help
BO-4	Prevent costly mistakes	Every ordering mistake defined in ZP-DOC-03 is detected in test, with no false "safe" claims
BO-5	Show every path	All seven route groups (state, private, diploma, professional, vocational, job exam, abroad) are covered for every A/L stream
BO-6	Operate at zero cost	Year-one hosting and inference stay within the Cloudflare free plan and free AI tiers

2.4 Market size

Table 4: Market sizing from the Gate 1 deck

Measure	Estimate	Basis
Total addressable market	about 275,000 students a year	All A/L candidates face an after-results choice (269,613 sat in 2023; 281,810 in 2025) [1]
Serviceable available market	about 105,000	Students who qualify (176,527 in 2025) × about 60% who use the internet [5]
Serviceable obtainable market (year one)	about 10,000	Tuition teachers, A/L Facebook groups, Telegram channels and word of mouth; to be validated

2.5 Success metrics

- **Adoption:** unique students per admission cycle (target BO-1); returning visits after results day.
- **Task success:** share of usability-test participants who complete a checked preference list unaided (BO-3).
- **Trust:** reported mistakes per 1,000 facts, and median time to fix a reported mistake.
- **Accuracy:** share of AI-extracted rules on which the extractor and the independent checker agree; share approved without change by the curator.
- **Cost:** daily usage as a percentage of each free-plan limit (BO-6).

2.6 Vision statement

> **For** Sri Lankan students between A/L results and their next step, **who** must choose among a state university, private degrees, job examinations and other routes within weeks and mostly on hearsay, **ZedPath is** a free, mobile-first guide **that** shows every path their own results can reach, with dated deadlines and answers that cite the official source, in Sinhala, Tamil and English. **Unlike** cut-off lookup sites and social-media groups, **ZedPath** checks subject and grade rules, preference order and deadlines, covers routes outside the UGC, and never states a fact without its source.

2.7 Business risks

Table 5: Business risks, their likelihood and impact, and mitigations

ID	Risk	Likelihood	Impact	Mitigation
RK-01	A wrong rule or cut-off leads a student to a costly decision	Medium	High	Every fact sourced and dated; two-model check; human review of uncertain facts; one-tap mistake reports; wording never promises admission
RK-02	Official sources change format or are published late	High	Medium	Source registry with fingerprints; extraction re-run per edition; *Revised (CR-002, CR-006):* a date not yet announced is shown as *Not announced yet* with the official rule and page, never estimated, and Journey steps cannot be ticked before it

ID	Risk	Likelihood	Impact	Mitigation
RK-03	Traffic spike on results day exhausts free-plan limits	Medium	High	Static front end served free; API responses cached; per-client rate limits; budget in ZP-DOC-06
RK-04	AI answers are wrong or poorly translated in Sinhala or Tamil	Medium	High	Answers grounded only in stored sources; refuse when unsupported; native-speaker review of interface text
RK-05	ZedPath is mistaken for an official UGC service	Low	Medium	Clear independence notice on every page; no UGC branding
RK-06	A single developer cannot finish R1 by 11 October 2026	Medium	High	Walking skeleton first; MoSCoW release plan; AI-assisted development; R2 holds deferrable stories
RK-07	Personal data of students (many are minors) is exposed	Low	High	*Revised (CR-001, CR-003):* only the details needed, each with its stated purpose and recorded consent (a parent or guardian agrees under 18); contact details kept apart and never shown to other students; passwords never leave the phone; secrets stored only as hashes; everything deleted with the account; photos deleted after reading. (v1.0: no account to explore; profile kept on the device.)
RK-08	*New (CR-002, CR-005).* Students, many of them minors, are harassed, misled or exposed in groups	Medium	High	Automatic moderation of every message, failing closed; rules accepted before taking part; phone numbers and emails refused; report (three reports hide a message) and block; rate limits; usernames and preset avatars only; seniors verified by the product owner through invite codes

2.8 Assumptions and dependencies

Table 6: Assumptions (AS) and dependencies (DE)

ID	Statement
AS-1	The UGC continues to publish its admission handbook and cut-off tables as public documents each year
AS-2	Government Gazette issues remain publicly available online each week
AS-3	Target students have a smartphone with a modern browser and some mobile data
AS-4	Cloudflare free-plan limits remain as verified on 5 October 2026 (recorded in ZP-DOC-06)
DE-1	The UGC handbook and cut-off table for the 2025/26 intake (received 5 October 2026)
DE-2	Cut-off tables for at least three earlier intakes, needed for Safe/Likely/Reach bands (US-203)
DE-3	A free tier of the Google Gemini API, accessed through Cloudflare AI Gateway (to be verified in ZP-DOC-06)

3 Scope and limitations

3.1 Major features

Each feature corresponds to one epic in ZP-DOC-02.

Table 7: Major features

ID	Feature	Epic
FE-1	*Revised (CR-001, CR-003, CR-006).* A required account with stated purpose and consent, the same profile on every device, a recovery code, account deletion and a preset avatar; results entered once (typed, estimated or read from a photo), in three languages	EP1
FE-2	Eligible state-university courses with Safe/Likely/Reach bands, reasons for hidden courses, and sourced course pages; *(VR)* course search and Compare of up to three courses	EP2
FE-3	Preference-list builder that detects ordering mistakes and copies uni-codes; *(VR, CR-006)* select several courses at once	EP3
FE-4	*Revised (CR-002, CR-006).* Journey the student ticks within each step's window, with followed courses and notices, confirmed dates only (never estimated), aptitude tests, reminders on the phone or in a calendar file, saved on the account	EP4
FE-5	*Revised (CR-002, VR).* Every other route matched to the student's results, in six groups (the retry path was removed), with Gazette job-exam and intake notices checked on the phone under the Journey's *Open now* tab	EP5
FE-6	*Revised (CR-004).* Ask: cited answers in Sinhala, Tamil and English that refuse to guess, shown as a card with a verdict, sources, next steps and feedback	EP6
FE-7	*Revised (CR-002, CR-005, CR-006).* Family share card; moderated groups per target course and per A/L batch with verified seniors, rules accepted once, and a member list; a teacher view	EP7
FE-8	After selection: registration, Mahapola, gap-year plan, documents	EP8
FE-9	Data and trust: source registry, AI extraction with two-model check, review queue, mistake reports	EP9

3.2 Scope of initial and subsequent releases

Table 8: Release scope by feature (story counts and points from ZP-DOC-02 v2.0 draft; superseded and withdrawn stories excluded)

Feature	Release R1, Gate 3 (11 October 2026)	Release R2, after the buildathon
FE-1	All 11 stories: results, estimate, photo, language, account, recovery, delete, contact details and consent, avatar, subject search	None
FE-2	8 stories: eligibility, hidden courses, bands, course page, filters, dream course, compare, course search	Cost of living
FE-3	All 5 stories, including selecting several courses	None
FE-4	9 stories: Journey, confirmed dates, aptitude tests, reminders, ticking steps, reminder settings, following courses and notices, Journey on the account	Special intake
FE-5	Route matcher, route details, Gazette and intake notices	Recognition check, loan scheme, scholarships
FE-6	All 7 stories: Ask in three languages, citations, no predictions, actions, answer card, feedback, achievements in answers	None

Feature	Release R1, Gate 3 (11 October 2026)	Release R2, after the buildathon
FE-7	7 stories: family share card, verified seniors, groups, safety, group rules, group info	Teacher view
FE-8	None	All 4 stories
FE-9	All 7 stories	None
Total	**57 stories, 189 points**	**10 stories, 31 points**

v1.0 planned 35 stories (122 points) for R1 and 15 (54 points) for R2. The change requests added 19 stories, moved four from R2 to R1 (answer actions, senior answers, senior verification, groups), superseded "custom journey steps" (replaced by following courses and notices) and withdrew "what retrying would take" with the retry path.

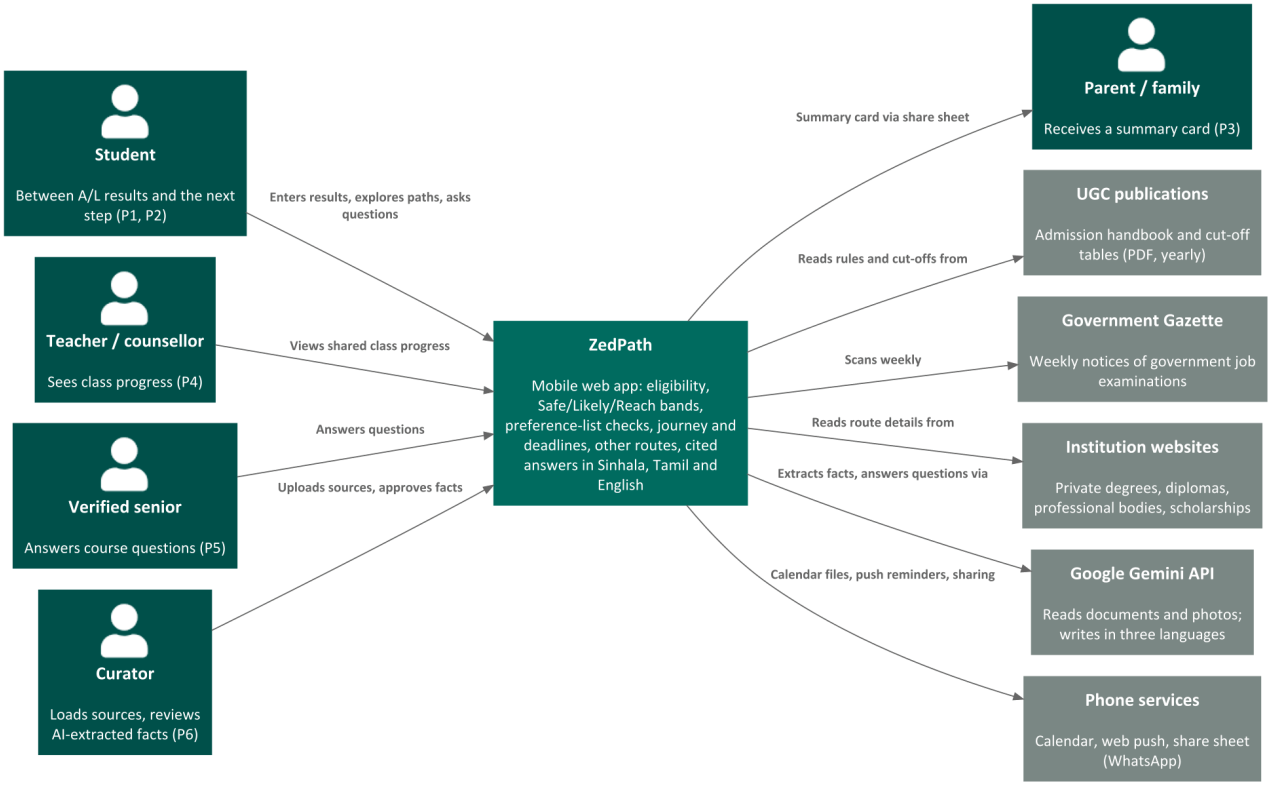


Figure 1: ZedPath system context (C4 model, level 1)

> **Note:** The context diagram shows the v1.0 baseline. Since CR-002 the student also uses an Android app that shows reminders through the phone's notification system, and group messages are screened by a moderation model on Cloudflare Workers AI; ZP-DOC-06 will show both.

3.3 Limitations and exclusions

ZedPath **will not**:

- submit applications or any data to the UGC or to any institution; it prepares the student to do so;
- predict future cut-offs or promise admission;
- accept payment for placement or ranking from any institution;
- send SMS, WhatsApp or email messages (not free on the chosen platform, and no sending domain or SMS service yet); *revised (CR-002):* reminders are notifications scheduled on the phone by the Android app, and calendar files;

- *revised (CR-003):* collect personal details beyond username, full name, email, mobile and date of birth, or keep any of them without a stated purpose and recorded consent; an account **is** required (v1.0: "require an account to explore");
- show a student's real name, contact details or photograph to other students; groups show usernames and preset avatars only (CR-006);
- keep results photos after they are read;
- cover postgraduate study or routes that do not start from A/L results;
- *revised (CR-002):* be a separate native app; it is a mobile web app that can be added to the home screen, and the Android app loads the same web app in its own window (v1.0: "be a native mobile app in R1").

4 Business context

4.1 Stakeholder profiles

Table 9: Stakeholders, their interests and constraints

Stakeholder	Main value to them	Attitude and interests	Constraints
Students (P1, P2)	Every reachable path, no costly mistakes	Anxious, time-pressed; trust peers more than institutions	Mobile data cost; language; low-end phones
Parents (P3)	A clear summary in their language	Want certainty about cost and recognition	Limited English; not app users
Teachers and counsellors (P4)	See who needs help before deadlines	Supportive; overloaded	No time for manual tracking
Verified seniors (P5)	A safe place to help juniors	Willing if protected from abuse	Must prove enrolment
Curator / product owner (P6)	Accurate data with little effort	Accountable for every published fact	One person in year one
UGC and Department of Examinations	Students who apply correctly	Publishers of the data, not partners	ZedPath must not appear official
Buildathon judges and mentors	Evidence of a real problem and a working, well-engineered solution	Assess Gates 1 to 3	Gate 3 due 11 October 2026

4.2 Project priorities

Table 10: Project priorities (constraint = fixed; driver = key success goal; degree of freedom = may be adjusted)

Dimension	Role	Explanation
Schedule	Constraint	Gate 3 submission is due on 11 October 2026
Cost	Constraint	Zero cash budget: Cloudflare free plan and free AI tiers only
Staff	Constraint	One developer (the product owner) with Claude as AI coding partner
Quality	Driver	Accuracy and sourcing of every fact matter more than feature count
Features	Degree of freedom	Could and Should stories move between R1 and R2 to protect quality and schedule

4.3 Deployment considerations

- ZedPath is a mobile web application delivered from Cloudflare's global network, so it needs no installation and loads quickly on Sri Lankan mobile networks. *Revised (CR-002):* an Android app loads the same web application and posts reminders as phone notifications, so every web deploy also updates the app.
- It must run entirely within the Cloudflare **free plan**; ZP-DOC-06 records the verified limits and the usage budget for each service.
- Interface text must render correctly in Sinhala and Tamil scripts on common Android phones.
- Official source documents are stored privately by ZedPath and cited, not redistributed.

Appendix A: Glossary

Table 11: Terms and abbreviations used in this document

Term	Meaning
A/L	G.C.E. Advanced Level examination
C4 model	A set of architecture diagrams at four levels; level 1, System Context, shows the system and its surroundings
Gazette	The Government Gazette of Sri Lanka, which announces government job examinations
CR	Change request: a change to the approved baseline, approved by the product owner
SAM, SOM, TAM	Serviceable available market, serviceable obtainable market, total addressable market
UGC	University Grants Commission of Sri Lanka
Z-score	A standardised score combining a student's three A/L subject results; used for university selection

Appendix B: References

Table 12: Sources referred to in this document

Ref	Source
[1]	Department of Examinations, Sri Lanka, *G.C.E. (A/L) Performance of Candidates*, 2023 and 2025
[2]	University Grants Commission, *University Admission 2025/2026*
[3]	National Education Commission, *Study on Career Guidance in General Education in Sri Lanka*, Research Series No. 08, 2014
[4]	Team Kestrel, *Gate 1 Problem and Proof*, IntelliCon '26 Buildathon, September 2026 (tool test of 27 September 2026)
[5]	DataReportal, *Digital 2026: Sri Lanka*
[6]	K. Wiegers and J. Beatty, *Software Requirements*, 3rd edition, Microsoft Press, 2013
[7]	G. Moore, *Crossing the Chasm*, HarperBusiness (vision statement template)